

WIPRO CAPSTONE PROJECT REPORT

Name	Telukuntla Sindhu
Superset Id	5481100
Organization	Wipro
Project Type	Capstone Project
Project Name	Automation Testing using Selenium Pytest
	Behave BDD Framework
Domain	Travel and Booking
Module Title	Ixigo Hotel Booking

SYNOPSIS

The Wipro Capstone Project focuses on automating the Ixigo Hotel Booking module using Selenium WebDriver with Python PyTest and Behave BDD frameworks. The automation framework validates the end-to-end hotel booking workflow including user login, destination selection, hotel search, filter application, room reservation, guest detail entry, and payment validation functionalities.

The framework was designed using the Page Object Model (POM) architecture to improve maintainability and reusability. Data-driven testing was implemented using CSV, while screenshots, logs, and Allure Reports were integrated for reporting and debugging.

Additionally, the Behave BDD (Behaviour Driven Development) framework was integrated using Gherkin syntax to create business-readable test scenarios. Feature files and step definition files were implemented for hotel booking workflows including hotel search, filter validation, guest detail validation, and payment validation. This improved readability, collaboration, and maintainability of automation test cases.

The project includes both positive and negative testing scenarios to validate application stability and functionality using industry-standard automation practices.

PROJECT DESCRIPTION

The automation framework automates the end-to-end workflow of the ixigo hotel booking module. The framework perform user login, destination selection, hotel search, filter application, room reservation, guest detail entry, and payment validation functionalities.

Separate page classes were implemented for Login, destination selection, hotel search, filter application, room reservation, guest detail entry, and payment validation modules using Selenium Python Page Object Model framework.

TEST STRATEGY

The project uses Selenium WebDriver with Python PyTest framework for automation testing. The framework supports:

- End-to-End Testing
- Positive Testing
- Negative Testing
- Data-Driven Testing

TOOLS USED

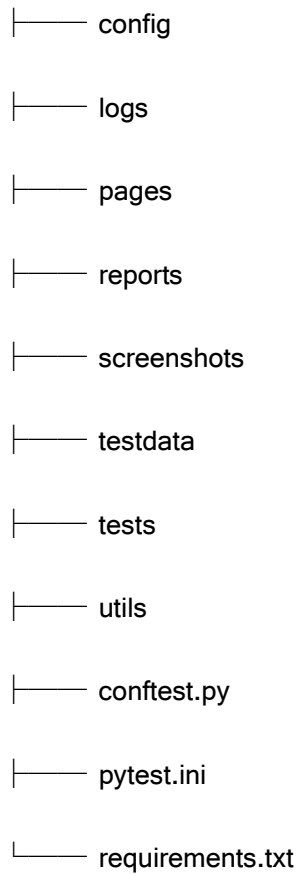
- Selenium WebDriver
- PyTest
- Behave BDD
- Gherkin
- Allure Reports

TEST ENVIRONMENT

Operating System	Windows 11
IDE	PyCharm Unified Product
Browser	Chrome or Edge
Programming Language	Python
Framework	Selenium WebDriver + PyTest
	Behave BDD
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports

FRAMEWORK ARCHITECTURE FOR SELENIUM PYTEST

IxigoHotels



TEST EXECUTION

The automation suite includes:

- 1 End-to-End Test Case
- 4 Positive Test Cases
- 2 Negative Test Cases

The End-to-End test validates the complete ixigo Hotel booking workflow from hotels to payment validation.

POSITIVE TEST CASES FOR SELENIUM PYTEST

Positive Test Case 1 – Hotel Search Validation

- Validates successful navigation to the ixigo hotels page
- Verifies destination search functionality
- Searches destination dynamically from search suggestions
- Checks successful hotel search result navigation

Positive Test Case 2 – Hotel Filter Selection Validation

- Validates hotel filter functionality
- Selects “Free Breakfast” filter
- Verifies successful filter application on hotel listings

Positive Test Case 3 – Guest Details Entry Validation

- Validates guest details form functionality
- Enters first name, last name, and email dynamically from CSV data
- Verifies successful guest detail submission
- Checks navigation to payment section

Positive Test Case 4 – Payment Details Validation

- Validates payment method selection functionality
- Selects Credit/Debit Card payment option
- Enters valid card details dynamically
- Verifies successful payment page validation workflow

NEGATIVE TEST CASES FOR SELENIUM PYTEST

Negative Test Case 1 – Invalid Card Number Validation

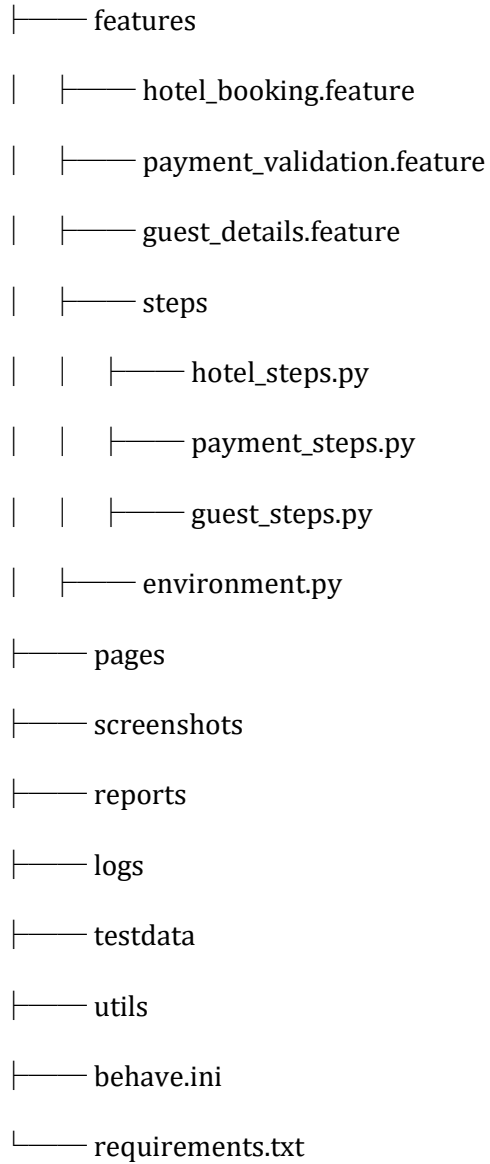
- Enters invalid card number with insufficient digits
- Clicks “Securely Pay” button
- Validates application error handling
- Verifies display of:
 -  “Please enter a valid card number”
- Ensures invalid card details are not accepted

Negative Test Case 2 – Empty Guest Details Validation

- Leaves mandatory guest details fields empty
- Attempts to proceed with booking workflow
- Validates mandatory field handling
- Verifies empty input validation functionality
- Ensures incomplete guest details are not accepted

FRAMEWORK ARCHITECTURE FOR BDD WITH BEHAVE

IxigoHotelsBDD



BDD TEST SCENARIOS

Feature: Ixigo Hotel Booking Functionality

Background

- User launches Ixigo application
- User submits valid mobile number for OTP
- User enters OTP manually
- Login flow continues after OTP validation

Positive BDD Scenarios

Scenario 1 – Search Hotels for Destination from Test Data

- User opens Hotels section
- User searches hotels using destination and dates from test data
- System displays hotel search results successfully

Scenario 2 – Apply Free Breakfast Filter on Hotel Results

- User opens Hotels section
- User searches hotels using destination and dates from test data
- User closes hotel results popup if present
- User applies “Free Breakfast” filter
- System validates successful filter application

Scenario 3 – Enter Guest Details on Booking Page

- User opens Hotels section
- User searches hotels using test data
- User selects first available hotel
- User reserves recommended room
- User enters guest details dynamically from test data
- Guest details should be filled successfully on booking page

Scenario 4 – Enter Card Details on Payment Page

- User completes hotel booking workflow
- User proceeds to payment section
- User enters payment card details from test data
- System validates payment details successfully

Negative BDD Scenarios

Scenario 5 – Validate Invalid Card Number Error

- User proceeds to payment section
- User enters invalid card details
- User clicks “Securely Pay” button
- System validates error handling
- “Invalid card number” validation message should be displayed

Scenario 6 – Validate Empty Guest Details Remain Blank

- User navigates to guest details page
- User leaves mandatory fields empty
- System validates mandatory field handling
- Empty guest detail fields should remain blank

Feature: Ixigo Hotel End To End Booking Functionality

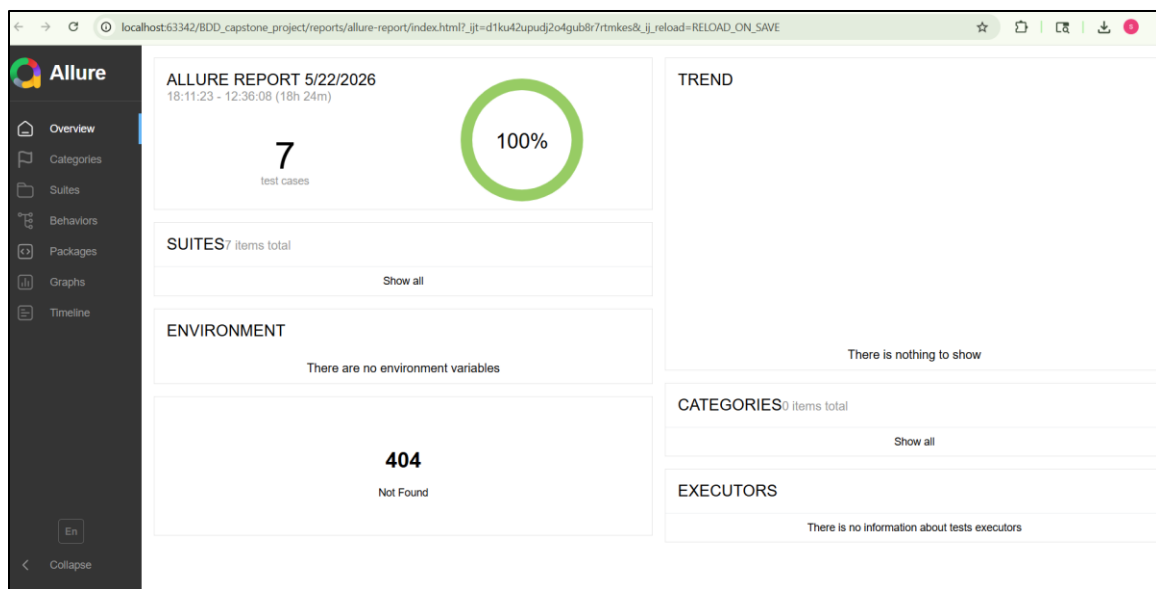
End-to-End BDD Scenario

Scenario – Complete End-to-End Hotel Booking Flow up to Payment

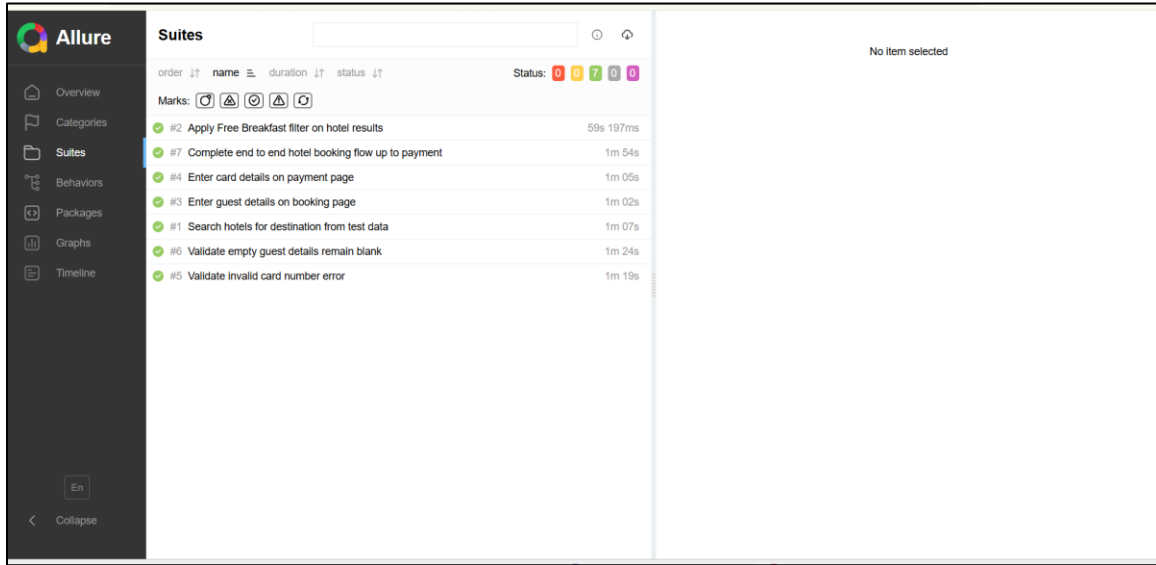
- User completes hotel search with Free Breakfast filter
- User completes hotel room booking with guest details
- User enters payment details using test data
- End-to-end hotel booking workflow should reach payment validation successfully

ALLURE REPORTS

- Integrated Allure Reports with PyTest
- Generates professional execution reports
- Displays passed and failed test status
- Captures screenshots during execution
- Maintains execution history
- Displays step-by-step execution flow
- Helps debugging and tracking failures
- Provides detailed execution summary



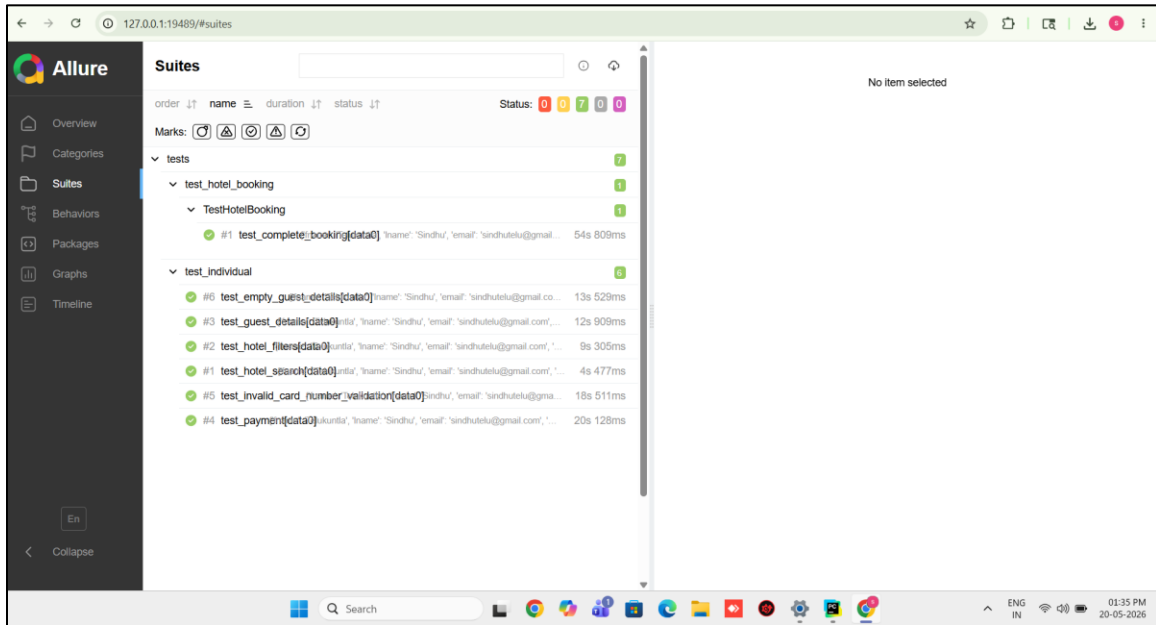
ALLURE REPORTS FOR SELENIUM WITH PYTEST



The screenshot shows the Allure report interface for Selenium with Pytest. The left sidebar contains navigation links: Overview, Categories, Suites (selected), Behaviors, Packages, Graphs, and Timeline. The main content area is titled 'Suites' and displays a table of test results. The table has columns for 'order', 'name', 'duration', and 'status'. The status column shows a color-coded indicator (green for pass, red for fail, yellow for skip, blue for pending, purple for error). The table lists 7 test items, all of which passed. The right sidebar shows 'No item selected'.

order	name	duration	status
#2	Apply Free Breakfast filter on hotel results	50s 197ms	Pass
#7	Complete end to end hotel booking flow up to payment	1m 54s	Pass
#4	Enter card details on payment page	1m 05s	Pass
#3	Enter guest details on booking page	1m 02s	Pass
#1	Search hotels for destination from test data	1m 07s	Pass
#6	Validate empty guest details remain blank	1m 24s	Pass
#5	Validate invalid card number error	1m 19s	Pass

ALLURE REPORTS FOR BDD WITH BEHAVE



The screenshot shows the Allure report interface for BDD with Behave. The left sidebar contains navigation links: Overview, Categories, Suites (selected), Behaviors, Packages, Graphs, and Timeline. The main content area is titled 'Suites' and displays a table of test results. The table has columns for 'order', 'name', 'duration', and 'status'. The status column shows a color-coded indicator (green for pass, red for fail, yellow for skip, blue for pending, purple for error). The table lists 7 test items, all of which passed. The right sidebar shows 'No item selected'.

order	name	duration	status
#1	test_complete_booking(data0) 'name': 'Sindhu', 'email': 'sindhuteku@gmail.com', '...	54s 809ms	Pass
#0	test_empty_guest_details(data0) 'name': 'Sindhu', 'email': 'sindhuteku@gmail.com', '...	13s 529ms	Pass
#3	test_guest_details(data0) 'name': 'Sindhu', 'email': 'sindhuteku@gmail.com', '...	12s 909ms	Pass
#2	test_hotel_filters(data0) 'name': 'Sindhu', 'email': 'sindhuteku@gmail.com', '...	9s 305ms	Pass
#1	test_hotel_search(data0) 'name': 'Sindhu', 'email': 'sindhuteku@gmail.com', '...	4s 477ms	Pass
#5	test_invalid_card_number_validation(data0) 'name': 'Sindhu', 'email': 'sindhuteku@gmail.com', '...	18s 511ms	Pass
#4	test_payment(data0) 'name': 'Sindhu', 'email': 'sindhuteku@gmail.com', '...	20s 128ms	Pass

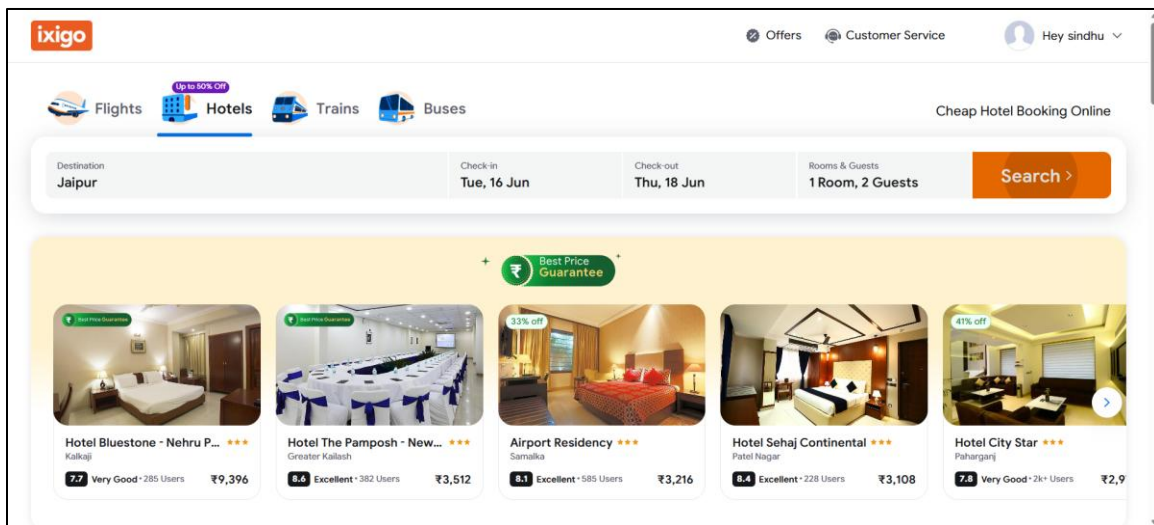
TEST OUTPUTS

The framework automatically captures screenshots during important execution stages including home page, hotel search page, filter selection page, booking details page, guest details page, and payments validation page.

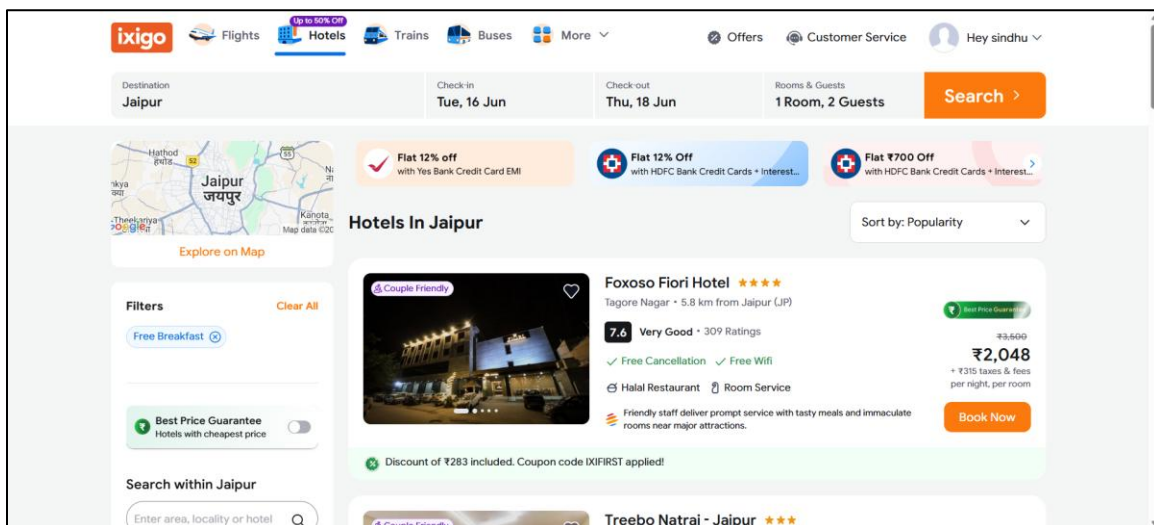
Detailed execution logs are generated using Python logging framework for debugging and execution tracking.

POSITIVE TEST OUTPUTS

HOTEL SEARCH VALIDATION



HOTEL FILTER SELECTION VALIDATION



GUEST DETAILS ENTRY VALIDATION

ixigo

Offers Customer Service Hey sindhu

Fare Summary

1 Room, 2 Nights ₹12,806

Discounts ₹6,735.30

Price After Discount ₹6,070.70

Taxes & Charges ₹960.30

Net Amount Payable ₹7,031

You'll earn ₹2000 Hotel Cash on this booking

Earned money will be credited within 24 hours of checkout date

Enter Your Details

☒ Mr. ☐ Mrs. ☐ Miss.

First Name
Telukuntla

Last Name
Sindhu

Email Address
sindhutel@gmail.com

Mobile Number
+91 6301249799

Nationality
India



Rajasthan Palace - A Heritage Boutique Hotel in Jaipur ★★★★★
3,Peelwa Garden,Moti Doongari Road,Raja Park,Near JLN Marg, Jaipur, 302004

1 x Deluxe Room 2 Guests

CHECK-IN
Tue, 16 Jun
1:00 PM

2 Nights

CHECK-OUT
Thu, 18 Jun
11:00 AM

Room Details

Offers For You

Have a promo code? Redeem here

☒ **IXIFIRST** ₹1037 off

Get ₹1037 off on your Hotel Booking
Know More

PAYMENT DETAILS VALIDATION

ixigo

09:58

Fare Summary

Amount To Be Paid ₹7,031

Yay! You saved ₹1037 on this booking



Rajasthan Palace - A Heritage Boutique Hotel in Jaipur
Adarsh Nagar, Jaipur

1 x Deluxe Room

CHECK-IN
Tue, 16 Jun
13:00

2 Nights

CHECK-OUT
Thu, 18 Jun
11:00

Guests

Telukuntla Sindhu +1 more

Recommended
Recently Used Methods

Pay via any UPI app
Scan and pay with UPI

Credit/Debit/ATM Card
VISA, Mastercard, Amex, RuPay & more

EMI
Credit & Debit Card EMI options

Wallets
Amazonpay

Net Banking
Select from list of banks

Add New Card

Card Number
4242 5252 6776 8998 VISA

Exp. Date (MM/YY)
12/29

CVV
123 456

☒ Save my card for future payments.

Securely Pay ₹7,031

100% Safe Payment Process

14

INVALID CARD NUMBER VALIDATION

EMPTY GUEST DETAILS VALIDATION

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TEST REPORT

- ✓ Total Tests Executed: 7
- ✓ Tests Passed: 7
- ✓ Tests Failed: 0
- ✓ Automation Accuracy: 100%

FINAL DELIVERABLES

- Selenium Automation Framework (Python)
- Behave BDD Feature Files
- Postman Collections + Newman Reports
- Test Report (PDF)
- GitHub Repository
- Demo Presentation

CONCLUSION

The Wipro Capstone Selenium Automation Framework successfully automated the ixigo Hotel Booking workflow using Selenium Python PyTest framework and BDD with behave. The project demonstrated real-time automation testing concepts including reusable framework design, Page Object Model architecture, data-driven testing, logging, screenshot capturing, and Allure reporting.